

Lecture 2a: Structure & Function of Cells in the Nervous System

Most cells in the body - including those in the Nervous System - share many basic features, including...

Soma = Cell body **Cytoplasm** = Fluid inside cell **Extracellular Fluid** = Fluid outside cell

Organelles in cells include...

Nucleus – where DNA (mostly instructions for building proteins) stored

Ribosomes - Site of Protein Production

- Proteins serve many functions (structure, transport, metabolism, membrane gates, etc)
- DNA sends “messenger RNA” from Nucleus, mRNA attaches to Ribosomes, instructs protein production

Mitochondria - Produce ATP (Adenosine triphosphate) whose breakdown frees energy to power cell functions

- Tend to cluster where “active” (energy-requiring) processes occur

Membrane - Lipids (fat molecules) w/hydrophobic cilia form double layered wall, 8nm thick

- Permeable to H₂O, O₂, CO₂, some fats; Generally impermeable to charged ions & larger molecules

Glial Cells – The non-neural cells of Nervous System; Many functions, but do NOT participate in Info Transfer

- “Glia” = “Glue”, hold the NS together, i.e. physically and chemically buffer and support the Neurons
- e.g. Astrocytes – Provide nutrients to Neurons, blood brain barrier, recycle NTs, remove waste, etc.
- e.g. Microglia - Proliferate in areas of brain damage, remove toxic materials (Like body’s immune system)
- e.g. Myelination - whole Schwann Cells wrap around axon in PNS, arms of an Oligodendrocyte in CNS
- e.g. Ependymal Cells – Line Ventricles, secrete Cerebral Spinal Fluid, beat cilia to circulate fluid
- e.g. Radial Glia - Guide migration and growth of Neurons during development (see below)
- Much smaller than most neurons (average 1/10), much more numerous (10X), so ~ 50% of brain by weight
- Unlike most Neurons, many Glial cells can regenerate (Runaway regeneration = “Glioma” brain tumor)

Neurons - Cells that are specialized for Information Transfer via modified 1) Processes and 2) Membrane

1) Processes = Elongated structures projecting from the “Soma” or cell body = Dendrites and Axon

Dendrites - Site of reception of incoming message

- From Greek “Dendron” = Tree, Usually many tapering, dividing branches, sometimes w/extra Spines
- **Receptor Sites** along surface interact w/molecules of **Neurotransmitter (NT)** from other Neurons
- Often have Ribosomes (for protein production, can be important for processing incoming message)

Axon - Site of release of outgoing message

- Each Neuron has, at most, one Axon, often long, non-tapering, may be Myelinated, end may branch
- Ends in **Presynaptic Terminals** (also called Terminal Buttons or End Bulbs) where **NT is released**
- Mitochondria in Terminals for energy-requiring processes like release/re-uptake of NT, ion pumps

2) **Membrane** As in *all* cells, lipid membrane generally impermeable to charged ions & larger molecules

- Neuron’s special **Selective Permeability** controls which chemicals enter/leave; affects electro-chemistry
- This done via gates (“channels”) that open or close to let chemicals (charged “ions”) pass through

The Nerve Impulse

To understand how Neurons “communicate” we first need to recognize that *Nature seeks a Balance . . .*

-i.e. Any Gradient (inequality) between the chemicals inside vs. outside cell will “seek” an equilibrium

Concentration Gradient - Molecules in area of greater concentration will **Diffuse** to area of lesser conc

Electrical Gradient - Positively charged particles will move away from other positive (& towards negative) and negative will move from negative (& towards positive) = **Electrostatic Pressure**

- In Neurons, the distribution of charged particles (**ions**, w/extra + proton or - electron) in/outside cells is controlled

- Recall how the Blood-Brain Barrier restricts what chemicals can move from bloodstream into brain

- **Membrane Potential** = Diff in charge in/outside cell, measured in millivolts (mV) using microelectrodes

- Key ions: Sodium **Na⁺** Potassium **K⁺** Calcium **Ca⁺⁺** Chloride **Cl⁻** as well as some charged Proteins

Resting Potential of most Neurons = **-70 mV** (less positive inside / more positive outside)

- Established in part by energy-requiring Sodium/Potassium Pump, actively transports **3 Na⁺ out** and **2 K⁺ in**
 - After transport, Na⁺ gates close, Na⁺ trapped outside (Extracellular fluid similar to seawater, w/NaCl salt)
 - K⁺ gates remain semi-open, K⁺ tends to leak out a bit
- Since Na⁺ now = 10:1 outside:inside, plus overall high quantity, Na⁺ “wants” to enter cell (to equalize both the Chemical and Electrical Gradients), but membrane is now impermeable to Na⁺
- Since K⁺ now = 1:10 outside:inside, plus lower overall quantity than Na⁺, K⁺ “wants” to exit cell to equalize the Chemical Gradient, but inhibited by Electrical Gradient (since outside more +)
- Cell’s negatively charged protein molecules are too large to leave, and closed Ca⁺⁺ gates keep Calcium out
 - Thus, Electrical Gradient also helps maintain more Cl⁻ outside (attracted to more + ions outside)
- Resting cell is thus **Polarized** = Large electro-chemical difference between inside/out = Ready to “fire”

Action Potential = Depolarization of Neuron = Cell “Fires” AKA “Spikes”

= Changes in membrane permeability of Axon, propagated via **Ionic Conduction**

- Triggered by NT from other Neuron, electrical stimulation, or other (see more on Synapse, below), typically...
 - Starting at **Axon Hillock** (where Axon joins Soma), **Voltage-activated Na⁺ gates open**
 - **Na⁺ rushes in**, reverses **local** polarization (depolarizes to +50mV)
- Na⁺ moving inside causes **adjacent Na⁺ voltage-activated gates to open**, & previously-open ones close
 - This depolarization sequence continues along Axon toward Terminal
- As previous **Na⁺ gates close**, local **K⁺ gates open wide**, **K⁺ leaves** (now per both chem & elec gradients)
 - Na⁺ gates continue to close, and K⁺ to open, following behind depolarization that is moving along Axon
- When depolarization reaches **Terminal**, **Ca⁺⁺ gates there open & Ca⁺⁺ enters cell**
 - Ca⁺⁺ influx leads to **Neurotransmitter (NT) release**
- As Membrane Potential again approaches more pos outside than in from K⁺ outflow, **K⁺ gates begin to close**
 - In time, **Sodium-Potassium Pump** actively restores Resting Potential (via 3Na⁺ out/2K⁺ in) => -70mV
 - And **Calcium Pump** actively rejects Ca⁺⁺ from terminal – Unlike passive ion flow above, **pumps req Energy**
- During **Refractory Period**, while cell is being re-polarized, it cannot fire (or resists firing)
 - Refractory Period also prevents impulse from being propagated back along Axon toward Soma by locking down gates behind, as impulse moves from hillock to terminal
- **All-or-None Law** = In a given cell, an Action Potential always has the same amplitude and velocity, regardless of the intensity of the stimulus that triggered it
 - Nonetheless, while the amplitude of the Spike (extent of depolarization) & amt of NT released is **fixed** the “message” such a cell can transmit can be varied through its...
Frequency of Firing (# spikes per sec) and **Pattern of Firing** (e.g. ||||| vs. || || || ||)

Myelination and Saltatory Conduction - Increases the speed of the propagation of an Action Potential

- **Glia cells** form insulating sheaths around some axons, with small gaps (**Nodes of Ranvier**) in between sheaths
 - **Oligodendrocytes** myelinate cells in Central NS (Brain and Spinal Cord), **Schwann Cells** in Peripheral NS
- **Electrical Conduction** (flow of electrons, like in an insulated wire) occurs along myelinated segment
 - But such signal **degrades** (weakens) as it moves, needs to be re-boostered, periodically, to original strength
- This occurs at **Nodes of Ranvier**, where electrical signal triggers the slower but stronger Ionic Conduction, which in turn triggers Electrical Conduction that moves rapidly under next sheath to next node, etc. etc.
- “**Saltatory**” = “Jumping” Nerve Impulse in effect “jumps” from node to node as it is propagated along axon
 - Increases overall speed of impulse from 1-10 m/sec to 100-120 m/sec!
- **MS (Multiple Sclerosis)** - Disease destroys myelin; In such un-insulated axons, electrical signal quickly degrades
 - Plus, since no Na⁺ gates under sheath, cell cannot resort to Ionic Conduction => cannot fire

Graded Potentials – Releasing NT from a Neuron does NOT always requires an Action Potential !

- e.g. Some **Receptor** cells (e.g. in Retina, Cochlea) react to outside stimulus (light, sound) w/graded potential
 - Loud sound >> large amounts of NT released, Soft sound >> little amount of NT released
- e.g. **Lateral Inhibitors** – cells that suppress neighboring cells so central cell’s message can get through
 - Often graded; The more/less excited the principal cell, the more/less inhibition to the neighbors
- e.g. Some Neurons are very small, have short or even no Axon or Dendrites
 - Called **Local Neurons**, these cells communicate only with immediately adjacent cells
 - In these Neurons, extremely rapid **Electrical Conduction** can cause NT release
 - Since cells so small, Electrical Conduction need not travel far, does not completely degrade
- Unlike Action Potentials, Graded Potentials can **vary in amplitude** in proportion to the input stimulus
 - i.e. React a lot to strong stimulus, less to weak one
- **NT Release** also tends to be **graded** (vs. fixed amounts typically released by Action Potential)